

Database Architecture & Schema (OPAY)

Offline UPI System | Data Persistence Strategy

1. Database Engine Strategy

The offline UPI architecture dictates a distributed database model across three unique environments, necessitating different persistence engines tailored to constraints and connectivity.

Environment	Engine	Primary Responsibility
iOS Client	SQLite (Core Data)	Immutable Offline Promise Queue, Encrypted Local History, Settings.
Android Gateway	SQLite (Room)	Webhook buffering, SMS state machine, localized routing logs.
Java Server	PostgreSQL / H2	Core Master Ledger, Double-spend prevention, Cryptographic Key Registry.

2. Server-Side Schema (Master Ledger)

The server must process concurrent asynchronous Webhook requests securely. Transaction isolation and indexing are critical for preventing replay attacks and race conditions.

Table: users

Maintains identity and cryptographic public keys for Secure Enclave validation.

Column	Data Type	Constraints & Indexes
id	UUID	PRIMARY KEY
phone_number	VARCHAR(20)	UNIQUE, B-Tree Index
public_key	TEXT	NOT NULL (Base64 encoded ECDSA Key)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Table: accounts

Manages the financial balances. Locked iteratively during transactions.

Column	Data Type	Constraints & Indexes
account_id	UUID	PRIMARY KEY
user_id	UUID	FOREIGN KEY (users.id)
balance	DECIMAL(12,2)	CHECK (balance >= 0)
status	VARCHAR(10)	'ACTIVE', 'FROZEN'

Table: transactions (Processed Ledger)

The authoritative record of completed actions. Used to block replay attacks.

Column	Data Type	Constraints & Indexes
txn_id	VARCHAR(32)	PRIMARY KEY , B-Tree/Hash Index
sender_phone	VARCHAR(20)	NOT NULL
receiver_phone	VARCHAR(20)	NOT NULL
amount	DECIMAL(12,2)	NOT NULL
device_timestamp	TIMESTAMP	Used for time-window validation
signature	TEXT	Hardware-backed validation signature
status	VARCHAR(15)	'SUCCESS', 'REJECTED'

3. Client-Side Schema (iOS Core Data)

Entity: OfflineQueue (The "Promise")

Acts as the append-only localized queue when the device lacks cellular connectivity.

Attribute	Type	Notes
local_uuid	UUID	PK

payload_blob	Binary Data	Pre-encrypted, Base85 compressed payload
prev_hash	String	SHA-256 of previous transaction in queue
dispatched	Boolean	Flag for SMS completion (Default: False)
creation_time	Date	Chronological sorting key for Priority Queue